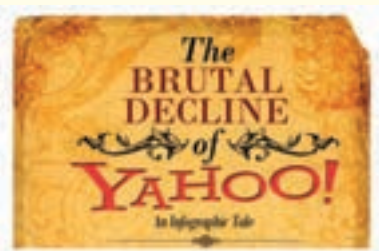


YAHOO! - THE RISE AND FALL OF AN INTERNET GIANT



<http://goo.gl/xd4ge>

MT. EVEREST GETS 3G MOBILE NETWORK

The highest point on planet Earth has now gone 3G. The mobile operator Ncell in Nepal has made this move to cater to the 30,000+ tourists that the peak attracts every year. The race is on: wonder who is going to become the mayor of the mountain by checking in to Foursquare first.

<http://goo.gl/Lv7vZ>

GROWING UP DIGITAL, WIRED FOR DISTRACTION

A new age look at how the next generation is growing up in a world that is now smaller than ever and where learning online is probably more productive than sitting in a classroom. With technology thrown into the mix of peer pressure and homework, how are the young-lings of the Google-age coping?

<http://goo.gl/ibZ9v>

and invalidates them. When a core reads data that another core has just written, the cache coherence protocol doesn't return the data until it finds the cache that holds the modified data, annotates that cache to indicate there is a copy of the data, and fetches the data to the reading core. These operations take about the same time as loading data from off-chip RAM (hundreds of cycles), so sharing mutable data can have a disproportionate effect on performance.

The conclusion, according to the MIT-news story, is that "[s]lightly rewriting the Linux code so that each core kept a local count, which was only occasionally synchronized with those of the other cores, greatly improved the system's overall performance." But will what works for modern-day Linux, and other operating systems, also work when the number of cores goes up even more? Past

this challenge. As this ConceivablyTech.com story says:

Intel has said in the past that, beyond 16 cores, it appears that much of the performance gain efficiency from the pure addition of cores may be gone and improving software that takes advantages of these cores may become much more important. That appears to be still true and we are actually seeing both processor makers buying themselves some time with the strategy of integrating GPUs into the processor package, which will provide a path to cost savings, but also a way to increase overall application speed as far as floating-point performance is concerned.

This has particularly applicability to Intel's recently announced next-generation processor microarchitecture, Sandy Bridge, which includes design features specifically aimed at helping it scale to machines with large numbers of cores. And AMD's own upcoming Fusion APUs, which seem intended to fully consummate the marriage that began when AMD acquired video powerhouse ATI a few years back, represent another step in the process. But will the software be ready?

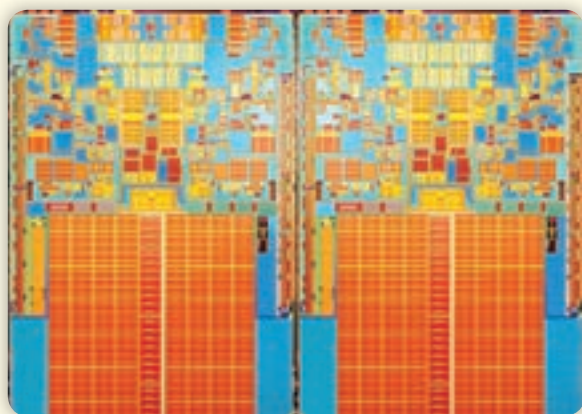
That's another question alto-

gether. Years after two- and four-core CPUs became commonplace in the market, even popular software apps were struggling to take full advantage of their capabilities. We've worked with some popular apps in our lab tests that profess multicore support, but don't show significant performance improvements when run on a six-core chip as compared to



a quad-core CPU with a similar clock speed. Things have gotten better in this area just over the last year and a half or so, but there's still a long way to go. So we're a little skeptical that software development will keep pace with the available hardware. (That it rarely has since the inception of the personal computing industry some 30-odd years ago makes it seem likely the trend will continue.)

Of course, there are lots of questions about this whole situation it's still too early to answer. Chief among them: Is 48 cores simply too many for any application, because the amount of time reassembling all the individual pieces of work will cancel out any benefit you may have gotten from doing them all at the same time? We have to hit that ceiling sometime—and if we're near it, how much of a difference can the software make, whether it's Linux, Windows, or anything else? In that case, we'll need even more inventive solutions to keep increasing performance. Let's hope that hardware and software companies alike are keeping that in mind as well, even as they're working to maximize the performance they can derive from systems with 48—or even six—cores.



48, according to one of the researchers, "new architectures and operating systems may become necessary."

Software and Hardware Solutions

It would seem that Intel and AMD are already working on separate ways to meet